Work Done

Exam code: 0625 & 0972

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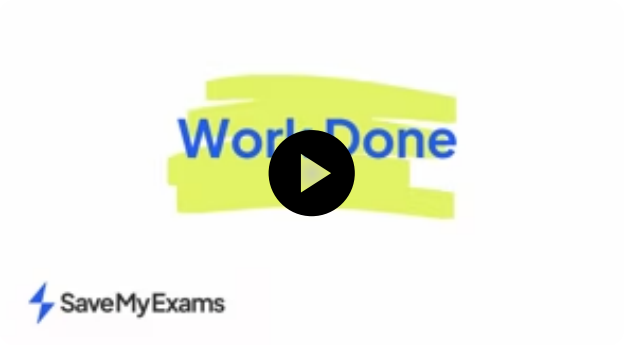

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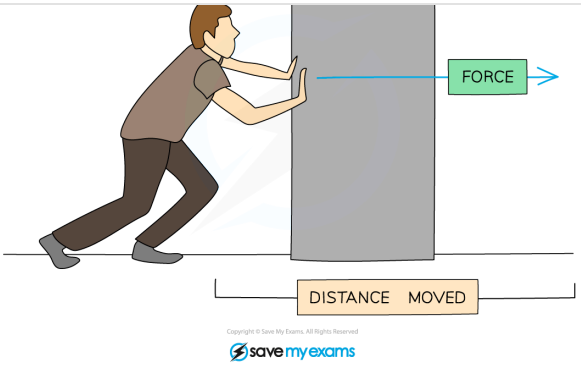
Work done & energy transfers

- **Mechanical work** is done when an object is moved over a **distance** by a **force** applied in the **direction** of its displacement
  - It is said that the **force does work** on the object
  - If a force is applied to an object but doesn't result in any movement, no work is done
- When **work** is done, **energy** is transferred
- Work done and energy transferred are **equivalent** quantities

Work done pushing a box

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Work is done when a force is used to move an object over a distance, and energy is transferred from the person to the box

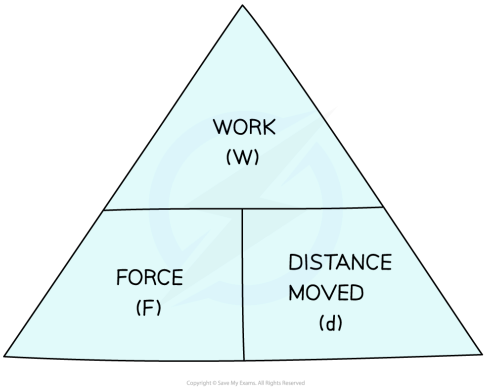
Work done equation

- The formula for work done is:

$$W = Fd = \Delta E$$

- Where:
  - W = work done, measured in newton-metres (N m)
  - F = force applied, measured in newtons (N)
  - d = distance in metres (m)
  - ΔE = energy transferred, measured in joules (J)

Formula triangle for work done, force and distance



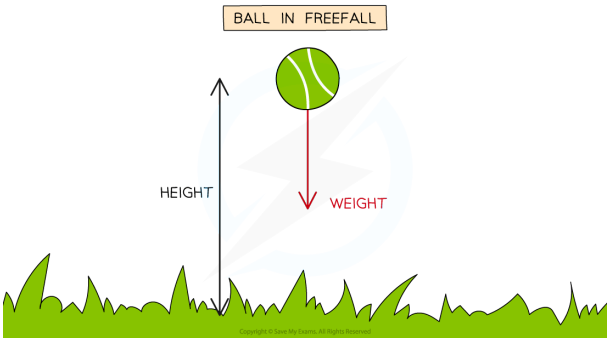
To use a formula triangle, simply cover up the quantity you wish calculate and the structure of the equation is revealed

- Using formula triangles is covered in more detail in the revision note on Speed & velocity
- Mechanical work done and electrical work done are equivalent to energy transferred

$$\text{work done} = \text{energy transferred}$$

- Therefore:

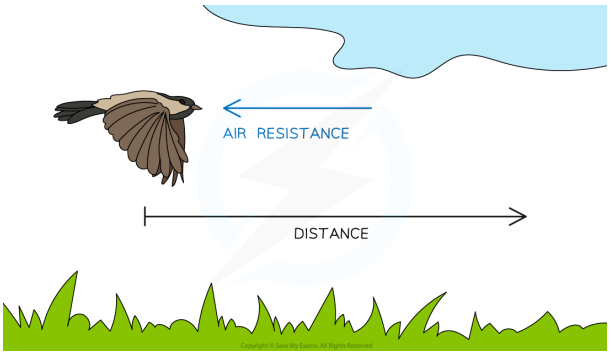
- Work is done on a ball when it is lifted to a height:
  - A force is required to lift the ball
  - Work is done against the weight force to lift the ball through the gravitational field
  - Energy is transferred as work in done



The weight on the ball produced by the gravitational field does work on the ball over a distance

- Work is done when a bird flies through the air
  - A force is required to overcome the drag force
  - Work is done against the drag force as the bird flies over a distance
  - Energy is transferred as work is done

Work done by a bird



The bird does work against air resistance (drag) as it flies through the air

Worked Example

A car moving at speed begins to apply the brakes. The brakes of the car apply a force of 500 N, which brings it to a stop after 23 m.

Calculate the work done by the brakes in stopping the car.

Answer:

Step 1: List the known quantities

- Distance, $d = 23\ \text{m}$
- Force, $F = 500\ \text{N}$

Step 2: Write out the work done equation

$$W = Fd$$

Step 3: Calculate the work done on the car by the brakes

$$W = 500 \times 23$$

$$W = 11\ 500\ \text{J}$$

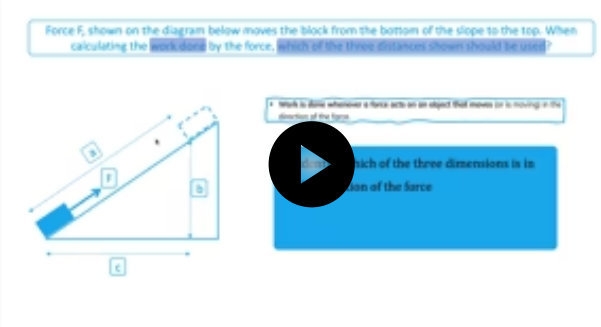
Step 4: Round to an appropriate number of significant figures

- The lowest number of significant figures in the input values in 1 s.f. (500 N)
- Therefore, the final answer must be rounded to 1 s.f.

$$W = 10\ 000\ \text{N}\ (1\ \text{s.f.})$$

Examiner Tips and Tricks

Remember to always convert the distance into **metres** and force into n**ewtons** so that the work done is in **joules** or **newton-metres**



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